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SARDAR PATEL UNIVERSITY

BCA 1st SEMESTER (Old Course) CBCS Examination-2023

US01BCA25 || Fundamental of Computer Organization

Date: 05/12/2023 Tuesday

Time: 10:00am to 01:00pm

Maximum Marks: 70

Q.1

Answer the following questions.

[10]

- 1 Where does a computer add and compare data?
(a) CPU (b) Hard disk (c) Floppy disk (d) Memory chip
- 2 Which parts of the computer perform arithmetic calculations?
(a) Registers (b) ALU (c) Logic bus (d) Input device
- 3 The brain of any computer system is
(a) Control Unit (b) Arithmetic Logic Unit
(c) Central Processing Unit (d) Storage Unit
- 4 ASCII equivalent of A is _____.
(a) 68 (b) 67 (c) 66 (d) 65
- 5 Extra bit added to a string of bits to detect errors is known as _____.
(a) Parity bit (b) Additional bit (c) Correction bit (d) updation bit
- 6 The maximum number stored in 8 bits using 1's complement is _____.
(a) -127 (b) 127 (c) -128 (d) 128
- 7 Array processor is referred as _____.
(a) SISD (b) MISD (c) SIMD (d) MIMD
- 8 In pipeline a single instruction is divided in _____ stages.
(a) 2 (b) 3 (c) 4 (d) 5
- 9 Which one is the input device _____
(a) Scanner (b) Printer (c) Plotter (d) none of these
- 10 Dot matrix printer is an example of _____.
(a) Serial printer (b) laser printer (c) drum printer (d) none of these

Q.2

Explain following in brief. (Attempt any Ten)

[20]

- 1 Define : Hardware with examples.
- 2 List applications of computer.
- 3 Define : Software with examples.
- 4 Explain 1's complement method with example.
- 5 Explain signed magnitude method with example.
- 6 Explain ASCII character code with example.
- 7 Explain RAM.
- 8 List steps of Instruction Execution cycle.
- 9 List stages of pipelining
- 10 What is Scanner?
- 11 What is mouse?
- 12 Give Two differences between input device and output device.

C.P.T.O.)

- Q-3 (A) Draw the Block diagram of Computer and explain its functions [05]
(B) Explain binary number system with example [05]

OR

- Q-3 (A) Explain first generations of computers [05]
(B) Explain hexadecimal number system with example [05]

- Q-4 (A) Explain Hamming code method with example [05]
(B) Explain CPU organization with diagram. [05]

OR

- Q-4 (A) Write a note on error detection and correction code with example [05]
(B) Explain UNICODE [05]

- Q-5 (A) Write a note on floppy disk. [05]
(B) Explain multifunctional unit [05]

OR

- Q-5 (A) Explain Flash drive. [05]
(B) What is ROM? [05]

- Q-6 Explain Keyboard with all kind of keys. [10]

OR

- Q-6 Explain impact printer. [10]

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